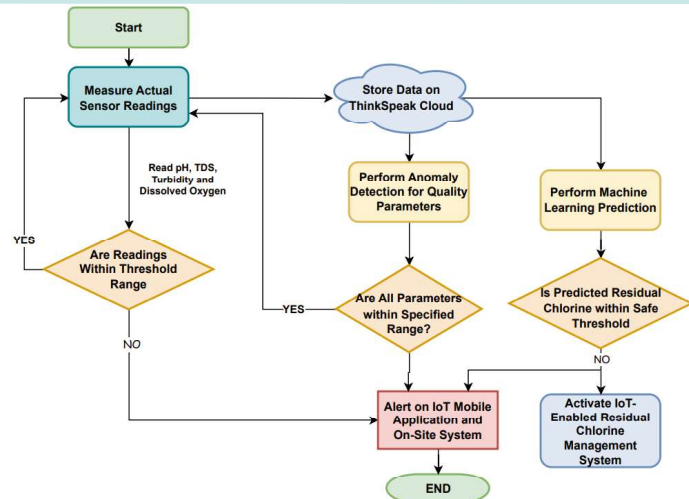


Intelligent Water Quality Monitoring and Residual Chlorine Prediction System Using IoT and AI

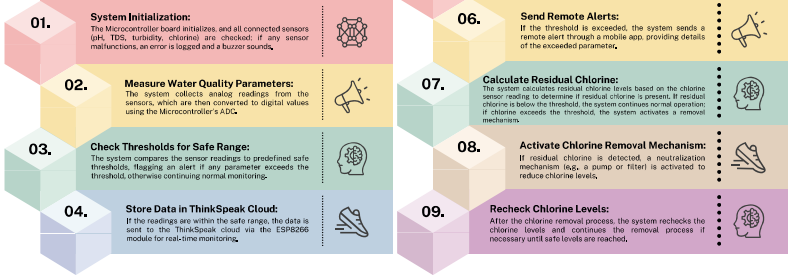
Abstract:

The "Intelligent Water Quality Monitoring and Residual Chlorine Prediction System Using IoT and AI" addresses the critical need for safe drinking water through continuous monitoring of key parameters such as Total Dissolved Solids (TDS), pH, turbidity, electrical conductivity, and chlorine levels. Utilizing IoT technology, the system collects real-time data and employs artificial intelligence to automatically regulate residual chlorine content, enhancing water quality with minimal user intervention. Coupled with a mobile application, it facilitates remote monitoring and early detection of water quality issues, thereby improving operational efficiency in water treatment processes and safeguarding public health.

Methodology:

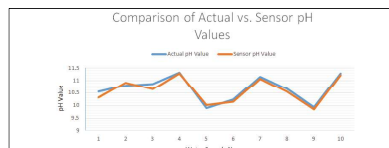
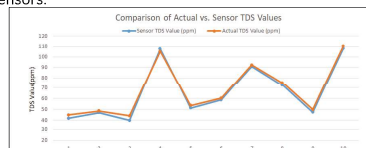


Procedure:



Results:

Initial readings from the pH and TDS sensors, collected from various rivers and water bodies around Pune District, have proven promising. The data recorded by our IoT-Enhanced Water Quality Monitoring and Residual Chlorine Management System closely aligns with the actual values, confirming the reliability and accuracy of our integrated sensors.



These results underscore the effectiveness of our system in providing real-time environmental monitoring. Further analysis and process enhancements will be conducted to continue improving the system's capabilities and precision in water quality management.

Introduction:

- Clean water is vital, but many regions struggle with pollution, industrialization, and climate change. Freshwater scarcity makes real-time monitoring critical for effective water management.
- The "Intelligent Water Quality Monitoring and Residual Chlorine Prediction System Using IoT and AI" tracks TDS, pH, turbidity, and chlorine levels and automates chlorine regulation and supports remote monitoring for efficiency.
- These systems provide accurate, on-site assessments, improving health protection where traditional testing falls short. Such innovations are key to safeguarding global clean water resources.

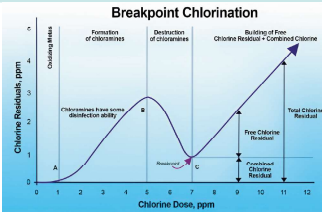
Need of Chlorine Monitoring:

Need for Chlorine:

- Chlorine is widely used in water treatment due to its strong disinfectant properties. It eliminates pathogens like bacteria and viruses, making drinking water safe.
- Chlorination is a low-cost, scalable solution that helps prevent waterborne diseases in public water systems.

Residual Chlorine Monitoring:

- Residual chlorine is the amount left after the disinfection process, ensuring ongoing water safety.
- Monitoring residual chlorine is crucial to prevent recontamination and maintain safe water quality. Continuous monitoring allows precise adjustments to maintain an optimal chlorine balance for safety.



Innovation :

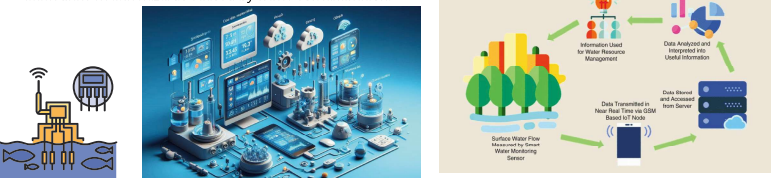
- Recent research has focused on error detection for water quality parameters, improving monitoring systems.
- Our research investigates using machine learning, specifically regression models, to predict residual chlorine concentration.
- Our project's USP is its unique focus on predictive chlorine management using machine learning and IoT.

Predictive Analysis of Residual Chlorine Concentration:

- We are exploring Regression Models such as Linear Regression, Random Forest Regression, and Support Vector Regression (SVR) for predictive analysis.
- Sensor data like pH, TDS, turbidity, chlorine and dissolved oxygen are used to learn the correlation with chlorine levels.
- The system accurately predicts when residual chlorine concentrations will exceed safe thresholds.
- IoT-enabled mobile applications provide alerts to users in advance of threshold breaches.
- The system triggers the chlorine removal mechanism, ensuring water safety and timely interventions.

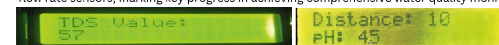
Anomaly Detection for Contamination :

- Benefit:** Machine learning models can automatically detect unusual patterns in real-time data (e.g., spikes in turbidity or pH) that indicate contamination events. This enables rapid response and containment of pollution before it spreads.
- Social Impact:** It significantly reduces the response time to water quality emergencies, protecting populations from immediate health hazards caused by water contamination.



Setup:

- The system now integrates sensors for Total Dissolved Solids (TDS) and pH, enabling continuous monitoring of the water's chemical composition.
- Real-time tracking of water volume and flow rate has been established through water level and flow rate sensors, marking key progress in achieving comprehensive water quality monitoring.



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Conclusion:

The "Intelligent Water Quality Monitoring and Residual Chlorine Prediction System Using IoT and AI" presents a significant advancement in ensuring safe drinking water through continuous monitoring of critical parameters. By utilizing IoT technology, the system captures real-time data on total dissolved solids (TDS), pH, turbidity, electrical conductivity, and chlorine levels. Its automated regulation of residual chlorine enhances water quality while minimizing the need for manual intervention. The integration with a smartphone app facilitates remote monitoring, enabling prompt detection of water quality issues and enhancing operational efficiency in treatment processes.